PRIORITY INTERRUPT

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-In Interrupt Initiated I/O transfer, the first task of the interrupt system is to identify the source of the interrupt. In the case several sources requesting for service, the system should decide which device to serve first.

-Priority Interrupt is a system which establishes a priority over the devices to determine which condition is to be serviced.

-Higher priority is assigned to the devices which if delayed, could have serious consequences. (High speed devices i.e. Magnetic disk)

-Establishing the priority of simultaneous interrupts can be done by software or hardware.

-Software – polling procedure is used to assign the priority.

-When the interrupt comes from several sources, the control is changed to a common branch address.

-Program that takes care of interrupts begins at the branch address and polls the interrupting sources in sequence.

-The order in which they are tested determines the priority of each interrupt.

-The device which is tested first has the highest priority and if it has set a interrupt, then control branches to a service routine for this source.

-Disadvantage: If there are many interrupts the time required to poll them can exceed the time available to service the I/O device.

Hardware Priority Interrupt:

- A hardware unit which function as an overall manager, accepts interrupt requests from many sources, determines the highest priority source and issues an interrupt request to CPU.

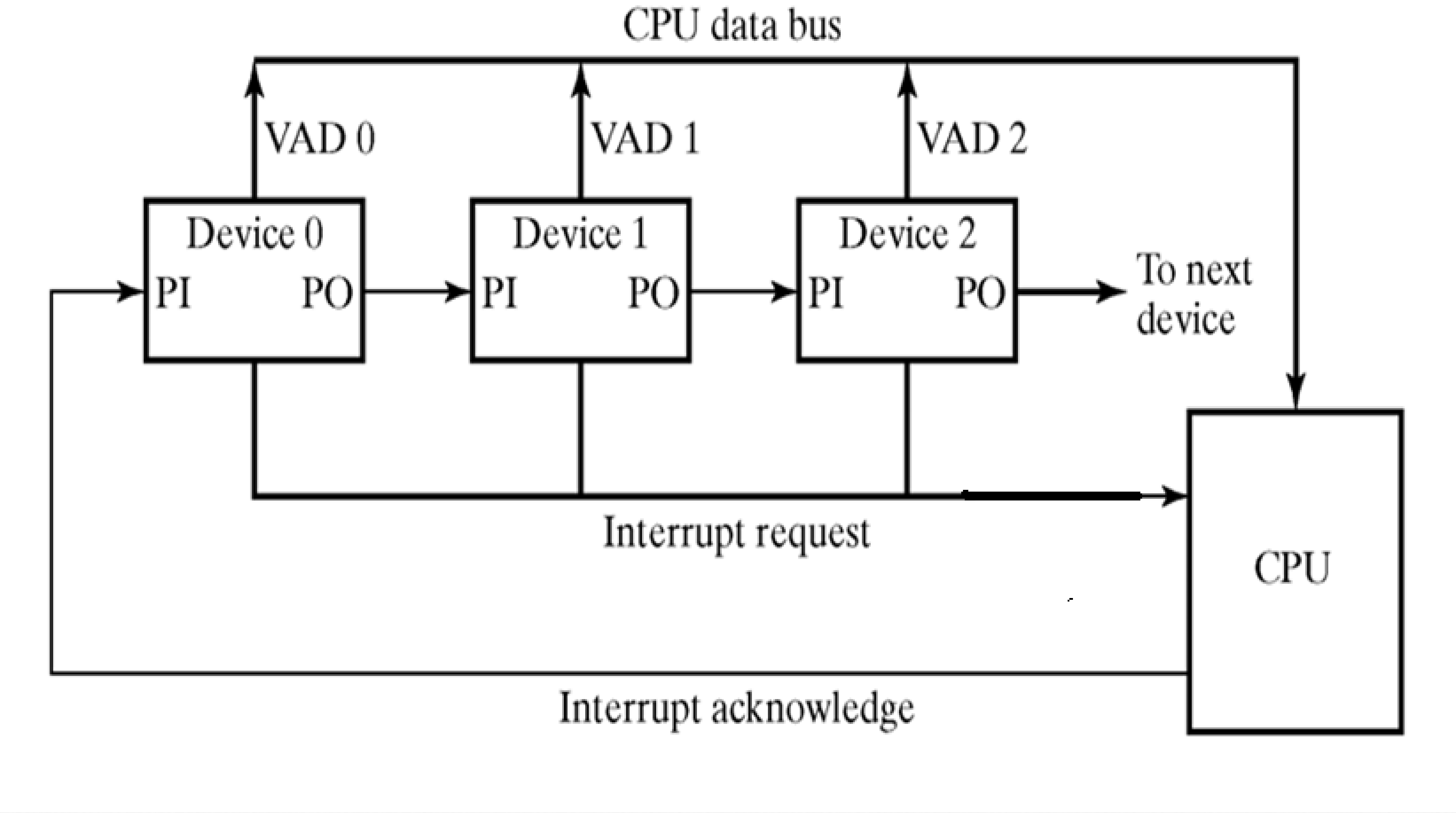
- Each interrupt source has its own interrupt vector to access its own service routine directly.

- The hardware priority interrupt function can be established in two ways:

1. Serial connection of interrupt Lines (Daisy- chaining method)

2. Parallel connection of interrupt Lines.

DAISY CHAIN PRIORITY INTERRUPT



- Serial connection of all devices that request an interrupt.

-Highest priority device is placed in the first position followed by the lower priority devices.

-Interrupt request line is common to all devices.

-If any device places interrupt signal, the interrupt request line is driven to low level state and enables an interrupt input to CPU.

-CPU responds by giving the interrupt acknowledge signal and the signal is received by device 1 at its priority in (PI).

-If device 1 has given the interrupt, it blocks the acknowledge signal by placing a 0 in PO output and inserts its own vector address (VAD) in to the data bus.

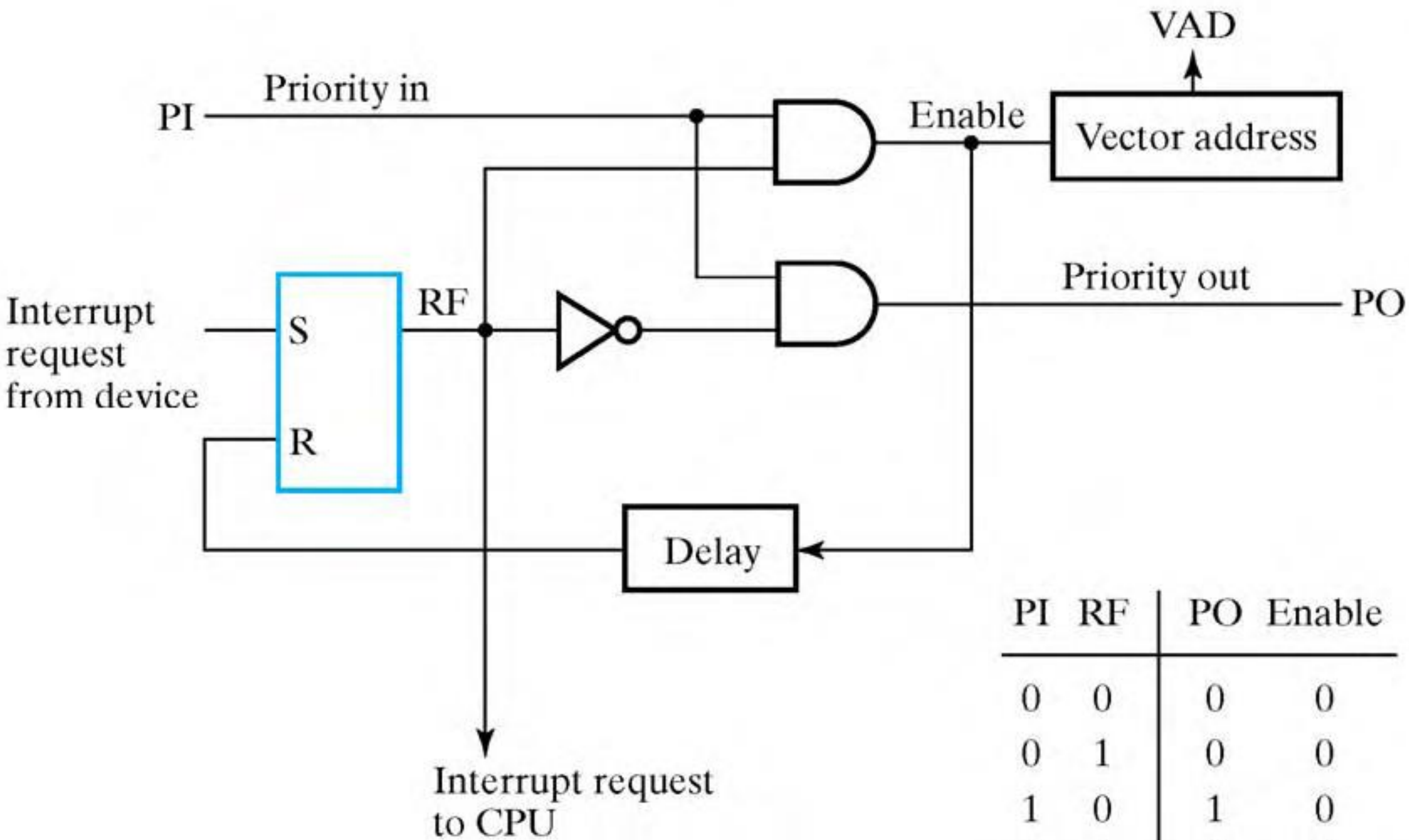
-If device 1 has not given the interrupt, it passes the acknowledge to the next device by placing 1 in PO.

-Three condition: 1. PI =0 (No acknowledge signal) PO= 0

2. PI =1 (device1 gives interrupt) PO =0

3. PI = 1 (device1 doesn't gives interrupt) PO =1

One stage of Daisy- chain priority arrangement



PI	RF	PO	Enable
0	0	0	0
0	1	0	0
1	0	1	0
1	1	0	1

PARALLEL PRIORITY INTERRUPT

-Interrupt Register: whose bits are set separately by the interrupt signal from each device and cleared by program instruction.

-Mask register: purpose is to control the status of each interrupt request. Its bit are set or reset by program instructions.

-The mask register can be programmed to disable lower priority interrupts while a higher priority device is being serviced and also provide a facility that allows a high priority device to interrupt the CPU while lower priority device is being serviced.

-Priority Logic:

-High speed device i.e. magnetic disk is given highest priority followed by printer, character reader and keyboard.

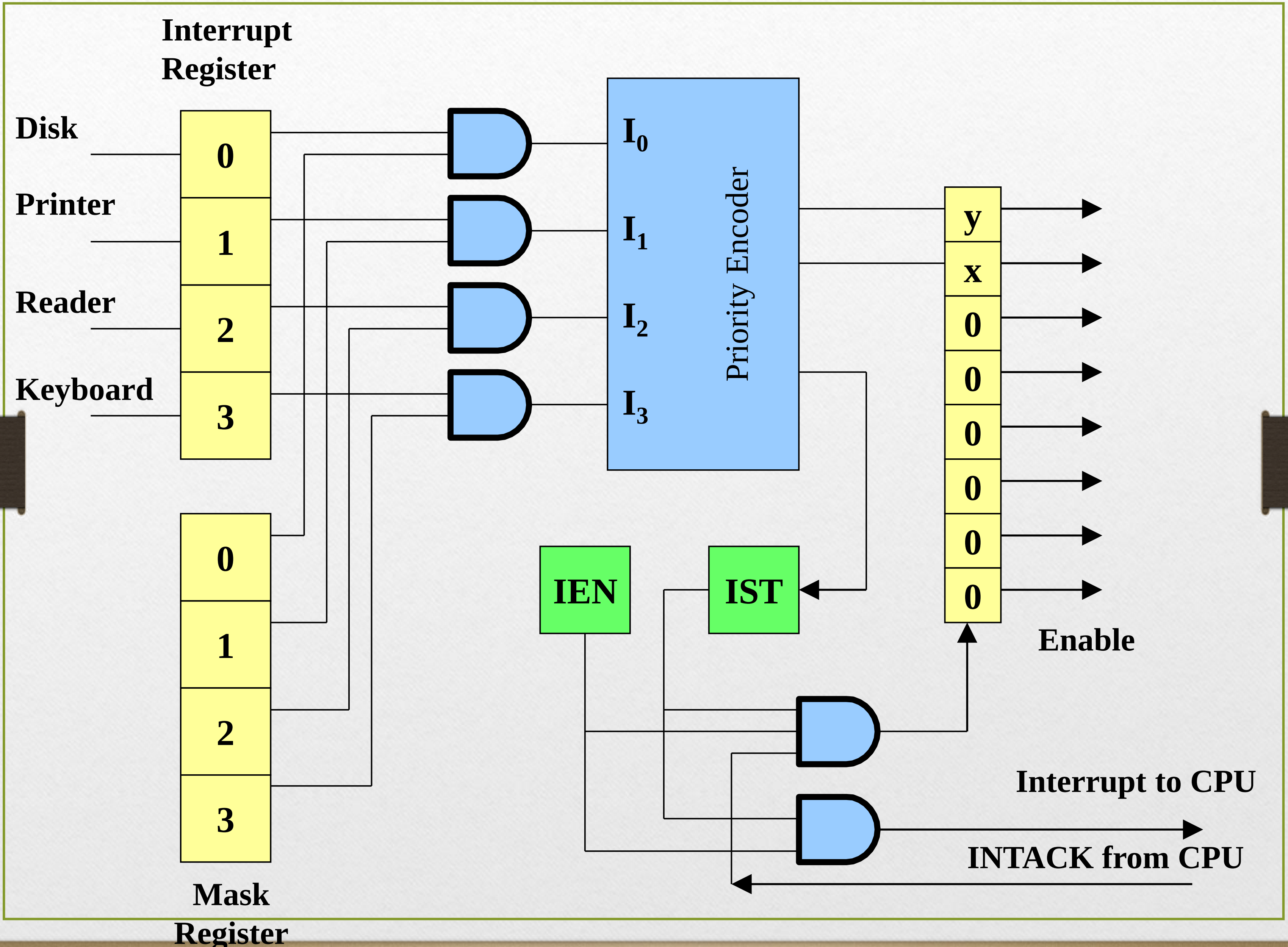
-The interrupt bit and its corresponding mask bit are applied to an AND gate to produce inputs to the priority encoder.

-Interrupt is recognized only if its corresponding mask bit is set to 1 by the program.

-Priority encoder generates two bits of the vector address which is transferred to the CPU.

-IST – Encoder sets this when one or more inputs are equal to 1.

-IEN – to provide overall control over the interrupt system.


Interrupt Register
Disk
Printer
Reader
Keyboard
0
1
2
3
0
1
2
3
Mask Register
I0
I1
I2
I3
Priority Encoder
y
x
0
0
0
0
0
0
Enable
IEN
IST
Interrupt to CPU
INTACK from CPU

INTACK: Signal from CPU which enables the bus buffers in the output register and a vector address VAD is placed into the data bus.

Priority Encoder:

-It is a circuit that implements the priority function.

-Logic of Priority encoder is such that if two or more inputs arrive at the same time, the input having higher priority will take precedence.

-Truth table:

$$X = I0'I1' \quad Y = I0'I1+I0'I2' \quad IST = I0 + I1 + I2 + I3$$

Inputs				outputs		
(I0	I1	I2	I3)	x	y	IST
1	x	x	x	0	0	1
0	1	x	x	0	1	1
0	0	1	x	1	0	1
0	0	0	1	1	1	1
0	0	0	0	x	x	0

- The Interrupt enable flip-flop (IEN) can be set or cleared by program instructions.
- A programmer can therefore allow interrupts (clear IEN) or disallow interrupts (set IEN)
- At the end of each instruction cycle the CPU checks IEN and IST. If either is equal to zero, control continues with the next instruction. If both = 1, the interrupt is handled.
- Interrupt micro-operations:
  - SP←SP – 1 (Decrement stack pointer)
  - M[SP] ← PC Push PC onto stack
  - INTACK ← 1 Enable interrupt acknowledge
  - PC ←VAD Transfer vector address to PC
  - IEN ←0 Disable further interrupts
  - Go to fetch next instruction